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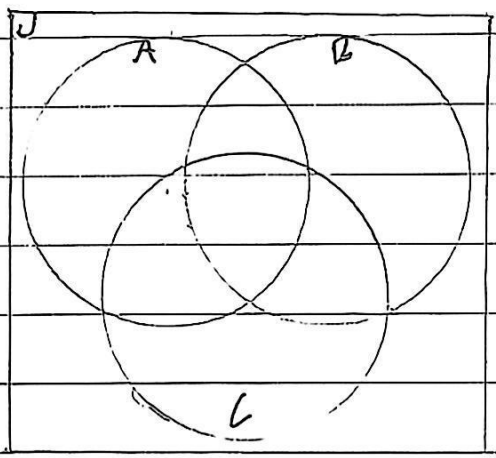
3. A : {17, 19, 21, 24, 27, 30, 32}

B : {20, 22, 24, 27, 29, 35}

C = {18, 24, 27, 28, 34, 37}

U : {17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37}



$|A \cup B \cup C| = \{17, 18, 19, 20, 21, 22, 24, 27, 28, 29, 30, 32, 34, 35, 37\}$

$|A \cap B \cap C| = \{24, 27\}$

$|(A \cup B) \cap C'| = \{17, 19, 20, 21, 22, 29, 30, 32, 35\}$

$|A \oplus B| = \{17, 19, 20, 21, 22, 29, 30, 32, 35\}$

b. $f(x) = 9x + 8$ | $g(x) = \frac{x^2}{(2+1)} + 1 = \frac{x^2}{8} + 1$

$(f \circ g)(x) = 9\left(\frac{x^2}{8} + 1\right) + 8 = \frac{9x^2}{8} + 9 + 8 = \frac{9x^2}{8} + 17$

$(g \circ f)(x) = \frac{(9x+8)^2}{8} + 1 = \frac{81x^2 + 144x + 64}{8} + 1 = \frac{81x^2}{8} + 18x + 8 + 1$

$= \frac{81x^2}{8} + 18x + 9$

$x = 7 + 2 = 9$

$(f \circ g)(9) = \frac{9 \cdot (9)^2}{8} + 17 = \frac{729}{8} + 17 = 91{,}125 + 17 = 108{,}125$

$(g \circ f)(9) = \frac{81(9)^2}{8} + 18(9) + 9 = \frac{6561}{8} + 162 + 9$

$= 820{,}125 + 171 = 991{,}125$

$(f \circ g)(x) \neq (g \circ f)(x)$

maka bilah komutatif

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d. $f(u) = 9u + 8$

$u = 9y + 8$

$u - 8 = 9y$

$y = \frac{u-8}{9}$

$f^{-1}(u) = \frac{u-8}{9}$

Pendapatan total

$k.100 + a + b$

$= 9.100 + 1 + 7$

$= Rp.908$

$f^{-1}(908) = \frac{908-8}{9} = \frac{900}{9} = 100$

e. Fungsi f merupakan bijektif karena setiap nilai dari domain memiliki nilai yang berbeda di kodomain dan setiap kodomain hanya memiliki satu himpunan di domain.

misal

$f(u_1) = f(u_2)$

$9u_1 + 8 = 9u_2 + 8$

$9u_1 = 9u_2$

$u_1 = u_2$

terbukti injektif karena menghasilkan output yang berbeda.

$y = 9u + 8$

$y - 8 = 9u$

$u = \frac{y-8}{9}$

terbukti surjektif karena jika memasukan nilai y rill akan menghasilkan nilai u rill

Fungsi g bukan bagian dari ketiganya

$u_1 = 4 \quad u_2 = -4$

$\frac{4^2}{8} + 1 = \frac{(-4)^2}{8} + 1$

$\frac{16}{8} + 1 = \frac{16}{8} + 1$

$3 = 3$

terbukti tidak injektif karena dari dua domain yang berbeda menghasilkan bagian kodomain yang sama

JOYKO 30 Lines, 6 mm

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Serta tidak surjektif karena kodomain memiliki nilai yang ≥ 0, untuk